

Multimodal Retrieval-Augmented Generation for Chart and Document Understanding: Architectures, Strategies, and Frontiers

1 Introduction

The paradigm of visuallanguage understanding has undergone a seismic shift from static, parameter-encapsulated knowledge to dynamic architectures leveraging non-parametric external memory. While early multimodal models relied on dense vector embeddings to align global image and text features [1], such monolithic approaches often struggle with the granularity and factual precision required for complex chart and document analysis. The emergence of Multimodal Retrieval-Augmented Generation (MMRAG) addresses these limitations by decoupling knowledge storage from reasoning, enabling models to retrieve specific visual evidence—such as chart axes, table cells, or document regions—to ground their generations. This transition is critical, as standard Vision-Language Models (VLMs) frequently hallucinate numerical values or misinterpret spatial relationships when relying solely on internal weights, a deficiency particularly acute in data-intensive domains [2].

Formally, MMRAG for charts and documents seeks to optimize the conditional probability $P(Y|Q, \mathcal{D})$ of generating a response Y given a query Q and a retrieved multimodal context $\mathcal{D} = \{I_k, T_k, L_k\}_{k=1}^N$, where I , T , and L represent visual patches, textual spans, and layout coordinates, respectively. Unlike unimodal retrieval systems that operate on homogeneous text corpora, MMRAG must navigate the heterogeneity of visually rich documents, where semantic meaning is inextricably linked to geometric arrangement. For instance, understanding a scientific figure requires not only recognizing objects but also interpreting the relational structures between subfigures and their captions, a challenge addressed by datasets like [3] and specialized frameworks such as [4]. The scope of this survey encompasses the full pipeline of this process: from structure-aware indexing strategies that preserve hierarchical document layouts [5] to generative synthesis mechanisms that fuse retrieved evidence without losing spatial coherence.

The distinctive challenges in this domain are manifold. First, preserving spatial coherence during retrieval is non-trivial; flattening a document into a sequence of tokens often destroys the two-dimensional dependencies essential for interpreting tables and infographics. Approaches like [6] and [7] highlight the necessity of modeling pairwise region relations and fine-grained word-region alignments to maintain structural integrity. Second, the semantic gap between natural language queries and symbolic graphical marks remains a significant hurdle. While models like [8] and [9] have advanced chart-specific pretraining, they often falter when faced with out-of-distribution chart types or complex multi-hop reasoning tasks requiring synthesis across disparate visual elements [10]. Furthermore, high-resolution inputs common in technical manuals and financial reports introduce computational bottlenecks that necessitate adaptive chunking and token merging strategies, as seen in recent efficient architectures [11].

Looking forward, the field is moving towards active, iterative retrieval frameworks where the generator dynamically determines the necessity and timing of evidence fetching, rather than relying on static, single-pass retrieval [12]. This evolution promises to mitigate the “over-reliance” on noisy retrieved contexts—a phenomenon observed in medical and legal domains where factual accuracy is paramount [13]. By integrating neuro-symbolic reasoning with robust retrieval mechanisms, future MMRAG systems aim to achieve a level of faithfulness and interpretability that bridges the gap between human expert analysis and automated understanding, ultimately transforming how we interact with the vast repositories of visual knowledge in science, finance, and beyond.

2 Multimodal Representation Learning and Indexing Strategies

2.1 Unified Multimodal Embedding Spaces for Visual-Textual-Layout Fusion

The construction of unified multimodal embedding spaces represents a paradigm shift from serializing document content into linear text streams toward jointly encoding pixel-level visual features, Optical Character Recognition (OCR) tokens, and geometric layout coordinates. Traditional retrieval pipelines often suffer from information loss when flattening two-dimensional structures, thereby severing the intrinsic semantic links between graphical marks and their textual labels. To mitigate this, contemporary architectures employ spatial-aware self-attention mechanisms that integrate 2D positional embeddings directly into the transformer backbone. For instance, [14] and [15] exemplify this approach by modeling the interaction among text, layout, and image modalities within a single framework, utilizing masked visual-language modeling and word-patch alignment objectives to capture cross-modal dependencies without relying on explicit bounding box annotations during inference. Similarly, [16] addresses the noise inherent in OCR-derived reading orders by introducing an Augmented XY Cut and dilated conditional position encoding, ensuring that the latent space preserves the robust spatial coherence essential for interpreting complex chart axes and document hierarchies.

Beyond spatial integration, effective fusion requires sophisticated alignment strategies to bridge the heterogeneity gap between dense visual signals and sparse textual semantics. Contrastive learning frameworks have emerged as a dominant strategy for aligning these modalities. [17] proposes aligning image and text representations before fusion via a contrastive loss, which facilitates grounded representation learning even in the presence of noisy web data. Extending this to the document domain, [8] introduces chart-specific pre-training tasks that explicitly model the structural relationships between data points and visual elements, enabling the model to reason over chart structures rather than merely recognizing visual patterns. Furthermore, generative objectives are increasingly utilized to refine these embeddings; [18] demonstrates that incorporating generative processes into cross-modal embedding allows models to learn not only global abstract features but also local grounded features, enhancing retrieval precision for complex content. This is complemented by approaches like [19], which leverage adversarial training to model the joint distribution of heterogeneous data, thereby learning a discriminative common representation that bridges the modality gap more effectively than deterministic mapping functions.

A critical challenge in this domain remains the handling of modality-specific characteristics and the imbalance of information density. While methods like [20] argue for constructing independent semantic spaces for each modality to preserve exclusive characteristics before adaptive fusion, others advocate for probabilistic embeddings to capture one-to-many correspondences inherent in image-caption pairs [21]. In the context of scientific and technical documents, where symbolic representations are nuanced, models such as [22] enhance layout awareness through reading order prediction and replaced regions prediction, while [23] bypasses traditional OCR entirely by translating academic documents directly into markup languages via a visual transformer, thus preserving mathematical and structural fidelity. Despite these advancements, significant gaps persist in generalizing across diverse document types and resolving fine-grained alignments in high-resolution settings. Future research must therefore focus on developing dynamic, query-aware embedding spaces that can adaptively weigh visual versus textual cues based on retrieval intent, potentially leveraging neuro-symbolic approaches to enforce strict geometric constraints within the latent manifold.

2.2 Structure-Aware Representation Learning via Graphs and Hierarchies

Bridging the gap between the unified latent spaces discussed previously and the flexible indexing strategies detailed subsequently, structure-aware representation learning explicitly models the complex topological, hierarchical, and relational dependencies inherent in visually rich documents and data visualizations. While prior approaches focused on embedding spatial coordinates and aligning modalities within a flat attention framework, they often struggle to preserve the explicit logical relationships critical for reasoning over complex charts, tables, and multi-page layouts. To address this limitation, recent advancements leverage Graph Neural Networks (GNNs) and tree-based encoders to construct heterogeneous graphs where nodes represent visual or textual elements—such as table cells, chart legends, or document sections—and edges encode spatial adjacency, reading order, or semantic dependencies. For instance, [24] introduced a graph convolution-based model that combines textual and visual information by training graph embeddings to summarize the context of text segments, significantly outperforming sequence-based baselines like BiLSTM-CRF. Similarly, [25] utilizes graph learning to fuse textual and visual features with global layout information, yielding richer semantic representations for key information extraction in complex document layouts.

The efficacy of graph-based modeling extends beyond static layout analysis to capturing fine-grained relational structures essential for precise multimodal retrieval. [26] proposes representing images and texts as visual and textual scene graphs, respectively, enabling cross-modal matching at both object and relationship levels. This approach refines node representations by aggregating neighborhood information, thereby facilitating more plausible similarity evaluations than coarse-grained methods. In the domain of diagrammatic reasoning, [27] introduces Diagram Parse Graphs (DPG) to model the syntactic structure of diagrams, while [28] employs multiplex graph networks to capture multiple relations between objects across diagram panels, achieving state-of-the-art accuracy in abstract reasoning tasks. Furthermore, [29] demonstrates the utility of dynamic graph attention networks in performing multi-step visual reasoning guided by linguistic structures, allowing for the precise localization of objects described in complex referring expressions.

Complementing these graph-based approaches, hierarchical encoding strategies address the multi-scale nature of documents and infographics, providing the structural backbone necessary for the multi-granularity indexing discussed in the next section. Frameworks like [30] utilize graph attention networks to simultaneously learn representations at document, paragraph, sentence, and token levels, effectively modeling dependencies between answers of varying granularities. This hierarchical perspective is crucial for long-context understanding, as evidenced by [31], which emphasizes unified structure learning across domains to boost the performance of Multimodal Large Language Models (MLLMs) on high-resolution images. Additionally, [32] adopts a coarse-grained approach that models contextualization between content blocks rather than individual words, avoiding excessive fragmentation while leveraging positional and visual information.

Despite these advances, challenges remain in balancing computational complexity with representational fidelity. While [33] highlights the difficulty of combining diverse inductive biases in heterogeneous graph datasets, emerging trends point towards hybrid architectures that integrate graph structures with transformer-based attention mechanisms. For example, [34] injects graph structures directly into the attention mechanism, allowing nodes to attend only to their neighborhoods within a multimodal framework. Future research must further explore dynamic graph construction for evolving data trends and develop efficient pruning strategies to handle the scalability of large-scale document corpora. Resolving these structural modeling challenges is a prerequisite for the effective decomposition of documents into flexible semantic units, a transition that enables the adaptive,

element-type-based chunking strategies explored in the following section.

2.3 Multi-Granularity Indexing and Adaptive Chunking Mechanisms

Effective multimodal retrieval-augmented generation (RAG) for chart and document understanding necessitates indexing strategies that transcend fixed-resolution segmentation, addressing the inherent heterogeneity of visual data structures. The core challenge lies in decomposing complex documents into flexible semantic units ranging from global page layouts to fine-grained data points, thereby enabling retrieval systems to match query intent with the optimal level of evidentiary detail. Traditional paragraph-level chunking often fails in visually rich domains, as it neglects the structural dependencies critical for interpreting tables and charts. Recent advancements advocate for element-type-based chunking, where documents are dissected according to their constituent semantic components—such as table rows, chart legends, or specific axis labels—rather than arbitrary text spans. This approach has been shown to significantly enhance retrieval accuracy in financial reporting by preserving the contextual integrity of structured data elements [35].

To support such flexible indexing, hierarchical representation learning is paramount. Frameworks like [30] utilize graph attention networks to simultaneously model documents at multiple levels of granularity, including tokens, sentences, paragraphs, and entire documents. This multi-layered encoding allows the system to capture dependencies between coarse-grained context and fine-grained entities, facilitating the extraction of both long-form answers and specific factual spans. Similarly, in the medical domain, [36] demonstrates the efficacy of aligning image and text representations across pathological region, instance, and disease levels, ensuring that retrieval mechanisms can bridge the semantic gap between high-level diagnostic queries and localized visual evidence.

Beyond static hierarchical indexing, emerging adaptive mechanisms dynamically select the appropriate granularity based on query complexity. This “Mix-of-Granularity” paradigm employs learnable routing strategies to direct queries toward the most informative index level, balancing the trade-off between broad semantic matching and precise numerical verification. For instance, [37] proposes learning discrete document identifiers (docids) that encapsulate semantic information, allowing generative models to retrieve relevant content by predicting these tokens end-to-end, effectively bypassing rigid chunk boundaries. Furthermore, [38] introduces a two-step process where large language models generate knowledge clues to guide multi-grained visual feature alignment, adapting the retrieval focus from global image semantics to specific object attributes.

The integration of extreme context compression further refines this adaptive landscape. Techniques such as [39] reinterpret dense retrieval embeddings as direct features within the language model’s latent space, effectively compressing entire document chunks into single tokens. This enables the system to handle massive corpora without sacrificing the ability to drill down into fine-grained details when necessary. Additionally, [40] enhances this capability by training models to dynamically focus on specific regions of interest across multi-page documents, utilizing hybrid visual vocabularies to switch between page-level overviews and region-specific OCR tasks.

Despite these advances, significant challenges remain in standardizing granularity definitions across diverse document types and managing the computational overhead of maintaining multi-scale indices. Future research must explore neuro-symbolic approaches that combine the flexibility of learned adaptive chunking with the rigor of structural constraints, ensuring that retrieval systems can robustly navigate the continuum from holistic document understanding to pixel-perfect data extraction.

2.4 Construction and Curation of Specialized Multimodal Knowledge Bases

Building upon the adaptive indexing and extreme context compression strategies outlined previously, the construction of specialized multimodal knowledge bases (MMKBs) represents the critical realization of these theoretical frameworks into dynamic, heterogeneous memory stores. While prior sections established how to decompose documents into flexible semantic units and compress them for efficient retrieval, this section addresses the paradigmatic shift from static, unimodal repositories to integrated systems capable of supporting complex reasoning over charts and documents. Unlike traditional retrieval systems that rely on lossy text extraction, modern MMKBs unify visual, textual, and structural modalities within cohesive indexing frameworks. A foundational challenge in this transition remains the automated extraction and normalization of chart-table data, where pipelines must accurately convert pixel-based representations into machine-readable structured formats. Approaches such as [41] and [42] demonstrate the efficacy of transforming visual elements into textual or structured representations, thereby enabling symbolic reasoning alongside semantic retrieval. However, purely structural conversion often fails to capture the nuanced perceptual trends inherent in data visualizations; consequently, recent frameworks like [8] and [43] advocate for pre-training strategies that align chart images directly with their underlying data tables, preserving the semantic fidelity required for accurate question answering [44].

To address the scarcity of high-quality, domain-specific annotated data necessary for populating these rich knowledge bases, synthetic data augmentation has emerged as a critical curation strategy. By leveraging large language models to generate diverse instruction-tuning datasets, researchers can populate knowledge bases with under-represented chart types and complex document layouts, effectively bridging the gap left by rigid structural schemas. For instance, [9] utilizes a multi-step generation process to create tabular data and corresponding chart figures, significantly enhancing the robustness of retrieval systems against distributional shifts. Similarly, [44] and [44] provide large-scale synthetic corpora that facilitate the training of models on specific reasoning tasks, such as identifying maxima or calculating area-under-curves, which are often absent in naturalistic datasets. Nevertheless, reliance on synthetic data introduces the risk of domain gap; thus, hybrid curation efforts like [45] and [46] emphasize the inclusion of natural, expert-verified charts from scientific literature to ensure real-world applicability and reduce hallucination risks during generation.

Beyond flat vector indices, the integration of Knowledge Graphs (KGs) with dense retrieval mechanisms—termed HybridRAG—offers a superior architecture for handling multi-hop queries across these heterogeneous document collections, directly addressing the limitations of simple semantic matching discussed in earlier granularity analyses. Traditional vector-based retrieval often struggles with the precise relational reasoning required in financial or scientific domains, whereas graph-based structures explicitly encode entity relationships and hierarchical dependencies. Works such as [27] and [47] illustrate how multimodal triplets can correlate visual objects with factual answers, bridging the semantic gap between imagery and external knowledge. Furthermore, [48] demonstrates that combining KG traversals with vector similarity search substantially improves retrieval accuracy in complex domains like finance, where inter-issue relations and intra-document structures are paramount. This synergy allows systems to perform not only semantic matching but also logical verification of retrieved evidence, as seen in medical applications where [49] and [50] utilize learned knowledge bases to ground radiology reports and electronic health records in verified clinical facts.

Despite these advancements, significant challenges remain in ensuring the scalability and quality control of MMKBs, particularly regarding the alignment between the compressed indices men-

tioned previously and the rich semantic stores constructed here. The “multi-granularity noisy correspondence” problem, where coarse-grained captions misalign with fine-grained visual details, necessitates sophisticated reranking and noise-injection training protocols [51]. Future directions must focus on developing self-correcting curation pipelines that continuously update knowledge bases with evolving document collections while maintaining strict provenance tracking. Ultimately, the efficacy of Multimodal Retrieval-Augmented Generation hinges on the construction of knowledge bases that are not merely large in scale but are structurally rich, semantically grounded, and rigorously curated to support faithful, evidence-based generation.

3 Advanced Retrieval Mechanisms and Context Alignment

3.1 Unified and Structure-Aware Cross-Modal Embedding Spaces

The construction of unified and structure-aware cross-modal embedding spaces constitutes the foundational layer for effective retrieval in chart and document understanding, necessitating a shift from flat image-caption pairing to representations that explicitly encode geometric topology and hierarchical dependencies. Traditional approaches often project heterogeneous modalities into a common latent space via shared projections, yet this can induce performance trade-offs between image-to-text and text-to-image retrieval tasks due to modality-specific information imbalances [52]. To mitigate such heterogeneity gaps, contemporary frameworks leverage generative adversarial mechanisms and contrastive objectives to model joint distributions, ensuring that the learned common representations preserve both intra-modality semantic consistency and inter-modality correlation [19; 53].

A critical advancement in this domain is the integration of layout and spatial geometry directly into the embedding process. Models such as [54; 54] employ spatial-aware self-attention and unified masking objectives to capture the intricate interactions among text, layout, and image modalities, thereby overcoming the limitations of sequential text serialization which often discards vital two-dimensional structural cues [45]. Similarly, [55] introduces robust layout-aware mechanisms via augmented XY cuts to handle variable reading orders, while [22] enhances pre-training with reading order prediction tasks to systematically mine layout-centered knowledge. These approaches collectively demonstrate that preserving the relative positional relationships among document blocks is essential for accurate semantic alignment, particularly in visually rich documents where text cannot be treated as a one-dimensional stream [24].

Beyond planar layouts, explicitly modeling relational structures through graph-based embeddings has emerged as a powerful paradigm for capturing fine-grained dependencies. By representing documents and charts as heterogeneous graphs where nodes denote visual or textual elements and edges encode spatial adjacency or logical dependency, Graph Neural Networks (GNNs) facilitate message passing that enriches local features with global contextual information [33]. For instance, [56] and [26] formulate retrieval as a graph matching problem, aligning object-relation structures in images with phrase structures in text to achieve superior fine-grained correspondence. This structural awareness is further refined in domain-specific applications; [36] illustrates the efficacy of aligning image and text at pathological region, instance, and disease levels simultaneously, while [57] utilizes multimodal graph contrastive learning to unify self-supervised pre-training across form document modalities.

Despite these advances, challenges persist in balancing modality-specific characteristics with unified representation learning. Methods like [20] argue against forcing a single common space, instead proposing independent semantic spaces fused via adaptive attention to respect the complementary

nature of visual and textual data. Furthermore, the emergence of generative retrieval and document screenshot embedding paradigms suggests a future direction where raw visual layouts are encoded directly without intermediate OCR bottlenecks, preserving holistic structural integrity [58; 18]. Future research must therefore focus on developing dynamic embedding spaces that can adaptively weigh structural versus semantic signals based on query intent, ensuring robustness against noisy layouts and varying granularities inherent in real-world chart and document corpora.

3.2 Hybrid, Sparse, and Generative Retrieval Paradigms

Building upon the foundational structure-aware embedding spaces discussed previously, the field is evolving from monolithic retrieval architectures toward hybrid, sparse, and generative frameworks to address the persistent heterogeneity in chart and document understanding. While the dense vector search methods established in prior sections excel at capturing high-level semantic concepts, they often struggle to localize precise numerical values or specific legend entries within complex visual layouts. Conversely, sparse lexical matching offers exact term localization but lacks the semantic depth required for reasoning over abstract visual trends. To bridge this dichotomy, recent advancements propose hybrid systems that dynamically fuse structural filtering, keyword matching, and dense semantic search. For instance, frameworks like [59] demonstrate that integrating knowledge graphs with vector indices significantly outperforms standalone approaches in financial domains by resolving complex multi-hop queries across heterogeneous document collections. Similarly, [60] introduces optimized retrievers that jointly handle tabular and textual evidence, mitigating the modality gap through mixed-modality negative sampling and synthetic pre-training.

A pivotal innovation within this landscape is the emergence of generative retrieval, which reframes the retrieval task as a sequence generation problem, moving beyond the static indexing limitations inherent in traditional embedding spaces. Instead of relying on explicit similarity scoring against an inverted index, generative models predict unique document identifiers or visual token sequences directly from the query. [37] proposes learning discrete document tokens via auto-encoding, enabling end-to-end optimization that captures semantic nuances often lost in rule-based identifiers. This approach is particularly potent for multimodal contexts; [61] assigns unique identifier strings to images, allowing multimodal large language models to “recall” visual content from parameters, thereby bypassing traditional indexing bottlenecks. Furthermore, [62] enhances this paradigm by constructing set-based lexical identifiers alongside sequential quantized representations, achieving substantial latency reductions while maintaining high precision.

The integration of sparse and dense signals is further refined through learnable projection mechanisms that complement these generative shifts. [63] illustrates that augmenting queries with heuristically generated contexts allows sparse representations (e.g., BM25) to rival dense retrievers, suggesting that hybridizing these modalities yields complementary benefits. In the visual domain, [58] advocates for treating document screenshots as unified inputs, leveraging vision-language models to encode layout and text simultaneously, thus preserving information often discarded during OCR-based sparse indexing. However, challenges remain in balancing the computational cost of generative decoding with the efficiency of sparse lookup. [64] addresses this by employing a cooperative retrieve-and-rerank strategy, where efficient bi-encoders perform initial candidate selection before cross-encoders refine the ranking, optimizing the trade-off between latency and accuracy.

Looking forward, the trajectory points toward fully unified generative frameworks capable of handling dynamic corpora, setting the stage for the iterative reasoning capabilities explored in subsequent sections. [65] highlights the necessity of incremental indexing strategies, such as product

quantization, to update document identifiers without catastrophic forgetting. As the field progresses, the convergence of sparse lexical precision, dense semantic robustness, and generative flexibility will be paramount for developing retrieval systems that can faithfully navigate the intricate spatial and symbolic structures inherent in scientific charts and technical documents, ultimately enabling the dynamic, multi-step reasoning processes required for complex query resolution.

3.3 Iterative, Active, and Intent-Aware Retrieval Frameworks

Moving beyond static, single-pass retrieval paradigms, iterative and active frameworks represent a paradigm shift toward dynamic, multi-step reasoning processes essential for complex chart and document understanding. These approaches treat retrieval not as a preliminary filtering step but as an integral, recurrent component of the generative loop, enabling systems to decompose ambiguous queries, assess knowledge gaps, and refine search criteria based on intermediate reasoning states. A critical advancement in this domain is the development of mechanisms for active retrieval timing and necessity judgement. Rather than indiscriminately querying external corpora, models like FLARE [66] utilize predictions of upcoming content to anticipate information needs, triggering retrieval only when generation confidence drops below a specified threshold. This strategy optimizes computational latency and mitigates noise injection, a principle further adapted for Large Vision-Language Models (LVLMs) in the ARA framework [12], which dissects image hierarchies to time retrieval during episodes of high visual uncertainty, effectively reducing hallucinations in chart interpretation.

For multi-hop reasoning tasks requiring the synthesis of disparate evidence from tables, text, and graphical marks, query decomposition has emerged as a dominant strategy. Frameworks such as Iter-RetGen [67] and SKURG [68] synergize retrieval and generation by using intermediate outputs to formulate subsequent sub-queries. This iterative refinement allows the system to navigate complex document topologies, retrieving specific data points or chart regions sequentially to construct a coherent reasoning chain. Similarly, the GenGround framework [69] alternates between generating hypothetical single-hop answers and grounding them in retrieved documents, amending errors through a feedback loop that progressively converges on the correct solution. In medical domains, i-MedRAG [70] demonstrates the efficacy of generating follow-up queries based on previous retrieval attempts, effectively simulating a clinician’s diagnostic reasoning process over complex visual and textual patient records.

Intent-aware retrieval further enhances these systems by aligning search strategies with nuanced user goals through query rewriting and expansion. The MIGRES paradigm [71] leverages the model’s ability to identify missing information to steer targeted knowledge retrieval, while RichRAG [72] employs sub-aspect exploration to handle broad, open-ended queries by retrieving diverse document facets. In interactive settings, systems like Ask&Confirm [73] actively seek discriminative visual details to resolve partial queries, shifting the burden from passive user feedback to active AI-driven clarification. Furthermore, instruction-tuned retrievers such as TART [74] and UniIR [75] demonstrate that explicitly encoding user intent via natural language instructions allows a single model to adapt to diverse retrieval tasks, from locating specific chart axes to summarizing entire document sections. Despite these advances, challenges remain in balancing the overhead of multiple retrieval steps with real-time performance requirements and ensuring that iterative loops do not propagate early retrieval errors. Future research must focus on developing robust convergence criteria for iterative processes and integrating neuro-symbolic reasoning to verify the logical consistency of aggregated multi-modal evidence.

3.4 Context Refinement, Reranking, and Noise Mitigation

Building upon the iterative retrieval frameworks and neuro-symbolic verification foundations established previously, the efficacy of Multimodal Retrieval-Augmented Generation (MMRAG) systems hinges critically on the subsequent post-retrieval refinement stage. Here, candidate contexts are rigorously evaluated to maximize semantic relevance and minimize the propagation of noise into the generative module. In chart and document understanding, raw retrieval results often contain visually distracting elements or semantically inconsistent fragments that, if unfiltered, can induce hallucinations or factual errors in the final synthesis. Consequently, advanced reranking mechanisms have emerged as an indispensable component, leveraging Multimodal Large Language Models (MLLMs) to perform fine-grained relevance scoring that transcends coarse vector similarity. Unlike traditional bi-encoder approaches that prioritize efficiency over precision, cross-encoder architectures facilitate deep interaction between the query and retrieved image-text pairs, enabling the model to discern subtle relational discrepancies in complex layouts [64]. By instruction-tuning MLLMs to act as discriminative rerankers, systems can effectively filter out candidates that fail to address specific numerical constraints or spatial relationships inherent in the user’s intent, thereby enhancing the fidelity of the context window [76].

A pivotal challenge in this domain is ensuring the robustness of retrieval against imperfect inputs, such as noisy OCR outputs or occluded graphical marks, which are prevalent in real-world document corpora. To mitigate these risks, recent methodologies incorporate noise-aware training objectives that expose rerankers to varying levels of visual and textual perturbation, fostering resilience against spurious correlations [77]. Furthermore, the integration of Knowledge Graphs (KGs) into the reranking pipeline offers a structural verification layer that complements dense vector retrieval, directly addressing the need for logical consistency highlighted in prior discussions. By intersecting retrieved candidates with KG traversals, systems can validate entity relationships and ensure contextual consistency, effectively resolving ambiguities that arise from heterogeneous data sources [59]. This hybrid approach, often termed GraphRAG, leverages topological information to prune irrelevant subgraphs and prioritize evidence that aligns with known domain logic, significantly reducing the likelihood of generating unsupported claims [78; 79].

Emerging trends also highlight the utility of iterative refinement loops, where the generation process itself informs subsequent reranking steps, creating a seamless feedback mechanism with the active retrieval strategies discussed earlier. Frameworks such as Iter-RetGen demonstrate that synergizing retrieval with generation in an iterative manner allows the model to dynamically reassess the quality of retrieved documents based on intermediate reasoning states, effectively correcting initial retrieval errors before final output synthesis [67]. Additionally, the concept of “generate-then-ground” introduces a paradigm where provisional answers are generated and then rigorously verified against retrieved evidence, amending predictions that contradict visual data [69]. Despite these advancements, trade-offs remain between computational latency and ranking accuracy, particularly when deploying heavy cross-encoders or graph-based verification on large-scale corpora. Future research must therefore focus on developing lightweight, adaptive reranking strategies that can dynamically balance depth of analysis with inference efficiency, ensuring scalable deployment in high-stakes domains like healthcare and finance where factual precision is paramount [80]. Ultimately, the evolution of context refinement mechanisms represents a critical frontier in bridging the gap between raw retrieval capability and trustworthy, grounded multimodal reasoning.

4 Generative Synthesis and Reasoning Over Retrieved Evidence

4.1 Multimodal Context Fusion and Architectural Integration

The effective synthesis of retrieved multimodal evidence hinges on architectural mechanisms that bridge the inherent modality gap between dense visual representations of charts and documents and the discrete token spaces of generative language models. This fusion process must transcend simple concatenation, ensuring that spatial layouts, numerical data points, and semantic text are coherently integrated to support complex reasoning tasks. Early approaches relied on cross-attention mechanisms to align visual regions with textual tokens, as seen in Dual Attention Networks which leverage fine-grained interplay between vision and language [81]. Similarly, Stacked Cross Attention frameworks have demonstrated the efficacy of inferring latent semantic alignments between image regions and sentence words to capture detailed interplay [82]. However, in the context of Retrieval-Augmented Generation (RAG), where context windows are constrained by retrieved evidence volume, direct attention over high-resolution document images often proves computationally prohibitive and semantically noisy.

To address these limitations, recent architectures have shifted towards parameter-efficient adapter mechanisms and structured serialization strategies. The LLaMA-Adapter V2 framework exemplifies this trend by employing an early fusion strategy where visual tokens are injected only into the initial layers of the Large Language Model (LLM), thereby minimizing interference with the model’s intrinsic linguistic knowledge while facilitating robust visual incorporation [83]. Complementing this, the xRAG approach pioneers extreme context compression by reinterpreting dense retrieval embeddings as direct features within the LLM’s latent space, effectively bypassing the need for verbose textual serialization and reducing computational overhead by orders of magnitude [39]. This modality fusion methodology allows the generator to operate on compressed document representations without sacrificing the fidelity required for precise data extraction.

Furthermore, preserving the structural integrity of retrieved documents—particularly the hierarchical relationships in tables and the geometric constraints of charts—remains a critical challenge. Traditional linearization often fails to capture the two-dimensional dependencies essential for accurate interpretation. To mitigate this, models like XYLayoutLM incorporate explicit 2D positional embeddings and layout-aware encodings to maintain spatial coherence during the fusion process [16]. Advances in tokenization, such as the SEED tokenizer, further enable the interchangeability of text and images within a unified autoregressive framework, allowing LLMs to “see” and “draw” by processing image tokens with causal dependencies aligned to linguistic prediction [55]. Additionally, the integration of visual prompting techniques, such as Set-of-Mark (SoM), enhances grounding by overlaying explicit markers on retrieved images, guiding the generator to specific regions of interest without requiring extensive retraining [84].

Despite these advancements, trade-offs persist between compression efficiency and information retention. While methods like H-Reducer in mPLUG-DocOwl 1.5 effectively merge horizontal patches to handle high-resolution inputs [31], excessive compression risks obscuring fine-grained numerical details crucial for chart analysis. Future architectures must therefore evolve towards dynamic, query-aware fusion mechanisms that adaptively select the granularity of visual evidence—ranging from global layout summaries to localized data points—based on the complexity of the reasoning task. The convergence of neuro-symbolic reasoning with these fused representations promises to further enhance the faithfulness of generated outputs, ensuring that statistical claims are rigorously grounded in the retrieved visual evidence.

4.2 Faithful Data-to-Text Synthesis and Statistical Grounding

Faithful data-to-text synthesis constitutes the critical final stage in multimodal retrieval-augmented generation, directly operationalizing the fused representations and dynamic fusion mechanisms discussed previously. While the preceding architectural innovations establish a robust foundation for integrating visual layouts and numerical data, the ultimate challenge lies in preserving the integrity of this retrieved evidence during natural language realization. The primary objective is to generate summaries and explanations that are not only fluent but strictly entailed by the underlying visual and tabular data, thereby mitigating the propensity of large language models to hallucinate statistical trends or fabricate values. Early approaches to chart summarization, such as those adapting transformer architectures for data-to-text tasks, demonstrated that while models could produce coherent narratives, they often failed to maintain factual consistency with complex visual patterns [41]. This limitation is particularly acute in open-ended question answering scenarios, where models struggle with the logical and arithmetic reasoning required to derive insights solely from chart images without explicit data tables [85].

To address these fidelity gaps, recent methodologies have shifted towards entailment-based decoding and reference-free verification mechanisms that build upon the grounding capabilities enabled by structured fusion. Frameworks like RAGAS provide a suite of metrics to evaluate answer faithfulness without ground-truth annotations, enabling the detection of claims that deviate from retrieved context [86]. Complementing this, specialized benchmarks such as MMHAL-BENCH focus on penalizing hallucinations in multimodal outputs, driving the development of factually augmented alignment strategies that incorporate image captions and ground-truth options into reward models [87]. Furthermore, the integration of visual grounding techniques, such as Set-of-Mark prompting, allows models to explicitly reference specific visual regions during generation, thereby anchoring textual claims to concrete graphical marks and reducing ambiguity [84].

Beyond simple extraction, advanced synthesis requires the articulation of perceptual features and cognitive phenomena, such as correlations, outliers, and long-term trends, pushing the boundaries of what static fusion can achieve. The VisText benchmark highlights the necessity of generating semantically rich captions that go beyond basic statistic reporting to describe the construction and perceptual salience of chart elements [46]. However, achieving this level of descriptive depth often exposes models to errors in trend interpretation, as seen in evaluations where state-of-the-art systems fail to correctly explain complex patterns despite high fluency scores [42]. To overcome this, emerging approaches like ChartThinker employ contextual chain-of-thought reasoning, synthesizing deep analysis by retrieving relevant strategies alongside data to improve logical coherence [88]. Similarly, ChartGemma addresses the disconnect between visual trends and underlying data by training directly on instruction-tuning data generated from chart images, ensuring that generated summaries capture both low-level visual information and high-level trends [89].

While these probabilistic and grounding-based approaches significantly enhance faithfulness, they often rely on the model’s internal capacity for arithmetic, which remains a bottleneck for rigorous quantitative analysis. Consequently, a promising direction involves hybrid neuro-symbolic frameworks that separate the linguistic generation from numerical verification, serving as a bridge to the program-aided paradigms explored next. By leveraging program-aided reasoning or enforcing strict structural constraints during decoding, systems can ensure that every quantitative claim is computationally derivable from the retrieved evidence. This is essential for domains requiring high precision, such as finance and healthcare, where even minor hallucinations can lead to significant misinterpretations. Future research must therefore focus on developing unified evaluation

standards that jointly assess linguistic quality and statistical fidelity, alongside training paradigms that inherently prioritize faithfulness over fluency. As the field evolves, the integration of active retrieval loops, where the generator dynamically queries for missing evidence to resolve uncertainties, will be paramount for achieving robust, trustworthy data-to-text synthesis in complex multimodal environments [66], setting the stage for the deterministic reasoning engines that follow.

4.3 Program-Aided and Neuro-Symbolic Reasoning Frameworks

The integration of program-aided and neuro-symbolic reasoning frameworks represents a paradigmatic shift in multimodal Retrieval-Augmented Generation (RAG) for chart and document understanding, addressing the inherent computational limitations of pure neural generation. While large multimodal models (LMMs) excel at semantic perception, they frequently falter in precise arithmetic and logical deduction required for complex data analysis. To mitigate this, recent architectures decouple the reasoning process into two distinct phases: a generative controller that synthesizes executable code or logical expressions, and an external deterministic interpreter that performs the actual calculation. This separation ensures mathematical fidelity by delegating numerical operations to reliable software engines rather than relying on the probabilistic token prediction of language models.

A foundational approach in this domain is the Program-of-Thoughts (PoT) strategy, which guides models to generate intermediate Python code to derive exact answers from retrieved chart data. [11] demonstrates the efficacy of this method by training compact models to emit Python scripts for numerical calculations, achieving state-of-the-art performance on benchmarks like ChartQA while significantly reducing parameter counts. Similarly, [90] illustrates that synthesizing reasoning traces via programmatic prompts enables smaller vision-language models to outperform significantly larger counterparts, effectively transferring the robust reasoning capabilities of Large Language Models (LLMs) to the visual domain. These frameworks treat the LMM not as a calculator, but as a semantic parser that translates visual queries into formal operations, thereby minimizing hallucination in multi-hop reasoning tasks.

However, the success of code generation is contingent upon accurate visual perception. Errors in interpreting graphical scales, axis units, or legend mappings can propagate into flawed program logic. To address this, visual perception alignment agents have been developed to explicitly map visual elements to semantic concepts before code synthesis. [91] pioneers a modality conversion approach where charts are translated into linearized tables, allowing LLMs to reason over structured text rather than raw pixels, though this risks losing nuanced visual trends. Conversely, [43] evaluates the cross-modal reasoning capability required to generate rendering code directly from images, highlighting the challenge of harmonizing visual understanding with syntactic correctness. Furthermore, [92] proposes a Description-Program-Reasoning chain specifically for graph theory problems, where generating a structural description precedes algorithmic reasoning, ensuring that the generated program aligns with the topological constraints of the visual input.

To further enhance reliability, self-correction loops have been integrated into these neuro-symbolic pipelines. These mechanisms allow the model to verify the output of executed code against the original visual evidence, iteratively refining the generated script if discrepancies arise. [93] exemplifies this compositional approach, where multiple pretrained models exchange information zero-shot to validate reasoning steps, effectively creating a feedback loop between semantic interpretation and symbolic execution. Additionally, [94] extends this concept by providing models with a visual scratchpad to draw auxiliary lines or marks, mirroring human problem-solving strategies to bridge

logical gaps before committing to a final program.

Despite these advancements, challenges remain in handling dynamic interactive visualizations and ensuring the generalizability of generated code across diverse chart styles. Future research directions point toward more tightly coupled neuro-symbolic architectures where the symbolic solver provides differentiable feedback to the neural encoder, enabling end-to-end optimization of both perception and reasoning modules. As the field evolves, the synthesis of executable logic with deep visual grounding will be critical for deploying trustworthy AI agents in high-stakes domains such as financial analysis and scientific discovery.

4.4 Iterative Refinement and Active Reasoning Loops

Building upon the neuro-symbolic foundations and self-correction loops established in the preceding section, iterative refinement and active reasoning loops represent a critical evolution in Multimodal Retrieval-Augmented Generation (MMRAG) for chart and document understanding. While previous architectures introduced static code execution and single-pass validation, these dynamic frameworks treat generation not as a terminal output but as an intermediate state that actively informs subsequent retrieval and evidence re-evaluation. This shift is particularly vital for resolving multi-hop queries in visually rich documents, where a single retrieval pass often fails to capture the distributed semantic and geometric dependencies required for complex reasoning [95]. By decomposing intricate questions into sequential sub-tasks, models can iteratively gather evidence from disparate chart regions or document pages, synthesizing a coherent response through a structured reasoning tree rather than a linear path [69].

A central mechanism in this domain is the active determination of retrieval necessity and timing, which extends the concept of self-correction into a proactive search strategy. Architectures such as FLARE (Forward-Looking Active REtrieval augmented generation) exemplify this by leveraging the model’s confidence in upcoming tokens to trigger targeted retrieval operations, effectively anticipating knowledge gaps before they manifest as hallucinations [66]. In the context of chart understanding, this translates to systems that dynamically query specific data points or axis labels only when the generative process encounters uncertainty regarding numerical trends or symbolic marks. This active loop is further enhanced by synergistic approaches like Iter-RetGen, where the output of one generation iteration serves as a refined query for the next, creating a feedback cycle that progressively sharpens the relevance of retrieved visual and textual contexts [67]. Such methods allow the model to correct initial misinterpretations of graphical scales or layout hierarchies, ensuring that the final synthesis is grounded in high-fidelity evidence.

Furthermore, the integration of self-memory mechanisms introduces a recursive dimension to iterative refinement, complementing the visual scratchpads discussed earlier. The SelfMem framework posits a duality where better generation prompts better memory retrieval, utilizing the model’s own high-quality outputs to expand the effective context window for subsequent reasoning steps [96]. In multimodal settings, this enables the construction of intermediate “scratchpads” where partial insights from chart elements are stored and re-indexed, facilitating the resolution of long-dependency relationships across multi-page financial reports or scientific figures [97]. However, these iterative loops introduce computational overhead and latency challenges. To mitigate this, corrective mechanisms like CRAG (Corrective Retrieval Augmented Generation) employ lightweight evaluators to assess the quality of retrieved documents at each step, triggering web searches or alternative retrieval strategies only when confidence scores fall below a threshold, thereby optimizing the trade-off between reasoning depth and efficiency [76].

Despite these advancements, significant challenges remain in stabilizing convergence within active reasoning loops, particularly when dealing with noisy OCR outputs or ambiguous visual layouts. Future research must focus on developing robust stopping criteria and adaptive granularity controls that allow models to switch between coarse-grained document overview and fine-grained pixel-level inspection dynamically. Additionally, extending these active loops to support human-in-the-loop refinement, where user feedback guides the retrieval trajectory in real-time, offers a promising direction for enhancing the interpretability and reliability of MMRAG systems in high-stakes domains [73]. Ultimately, the transition from passive context injection to active, iterative reasoning marks a fundamental step toward autonomous, trustworthy multimodal agents capable of deep, evidentiary synthesis, setting the stage for the specialized hallucination mitigation strategies explored in the following section.

4.5 Hallucination Mitigation and Factuality Calibration

Hallucination in multimodal retrieval-augmented generation (MMRAG) manifests as the synthesis of textual claims unsupported by retrieved visual evidence or the misalignment of numerical data with graphical marks. Mitigating this requires a dual strategy: training-free decoding interventions that enforce consistency during inference, and fine-tuning approaches that recalibrate the model’s reliance on parametric versus non-parametric knowledge. A primary avenue for training-free mitigation involves contrastive decoding mechanisms that penalize tokens diverging from visual constraints. For instance, frameworks leveraging visual self-consistency checks utilize reconstructed images or alternative descriptions to suppress hallucinated outputs, ensuring that generated text remains entailed by the source material [98]. Furthermore, the integration of active retrieval timing allows models to dynamically invoke external knowledge only during episodes of low certainty, thereby circumventing the injection of noise when internal parametric knowledge is sufficient [12]. This selective retrieval is critical, as indiscriminate augmentation can exacerbate hallucination degrees in large vision-language models (LVLMs) [12].

Beyond decoding constraints, calibrated retrieval dependence addresses the risk of over-reliance on potentially noisy retrieved contexts. In high-stakes domains such as healthcare, models must balance fidelity to retrieved evidence with robust internal medical knowledge. The RULE framework exemplifies this by introducing a provably effective strategy for controlling factuality risk through the calibrated selection of retrieved context counts, preventing errors where correct internal knowledge is overridden by irrelevant retrieved snippets [13]. Similarly, Corrective Retrieval Augmented Generation (CRAG) employs a lightweight retrieval evaluator to assess document quality, triggering large-scale web searches when local corpora yield sub-optimal results, thus enhancing robustness against retrieval failures [76]. These approaches highlight a shift from static retrieval pipelines to dynamic, confidence-aware systems that modulate the weight of external evidence based on uncertainty estimates.

Fine-tuning strategies further reinforce factuality by aligning generative objectives with factual verification signals. Factually Augmented RLHF adapts reinforcement learning from human feedback to the vision-language domain, where reward models are augmented with ground-truth image captions and multi-choice options to penalize hallucinations explicitly [87]. This method significantly improves performance on benchmarks designed to detect hallucinations, such as MMHAL-BENCH, by simulating human preferences for grounded responses. Additionally, noise-injected training protocols enhance generator robustness by exposing models to multi-granularity noisy correspondences during instruction tuning, enabling them to filter out distracting visual elements common in real-world documents [51]. In specialized domains like radiology, grounded report generation tasks

enforce localization of findings, ensuring that every generated sentence corresponds to a specific visual region, thereby providing verifiable traceability [99].

Despite these advancements, challenges persist in balancing the trade-off between strict faithfulness and generative fluency. Over-constraint can lead to overly conservative outputs that fail to synthesize high-level insights, while under-constraint risks factual fabrication. Future research must focus on developing neuro-symbolic verification modules that can formally prove the consistency of generated statements against retrieved structured data, moving beyond probabilistic alignment to deterministic factuality guarantees. Furthermore, the development of domain-specific ontologies for medical and legal RAG systems will be essential to enforce hard constraints on terminology and logical relations, ensuring that automated reasoning adheres to rigorous professional standards.

5 Evaluation Frameworks, Benchmarks, and Domain Applications

5.1 Benchmark Datasets for Multimodal Reasoning and Retrieval

The evaluation of Multimodal Retrieval-Augmented Generation (MMRAG) systems necessitates benchmarks that transcend simple image-caption matching, demanding rigorous assessment of multi-hop reasoning across heterogeneous visual-textual landscapes. The current landscape is characterized by a paradigmatic shift from synthetic, template-based queries to datasets derived from real-world scientific, financial, and medical corpora, which inherently possess complex layout structures and domain-specific semantics. Foundational efforts like [100] and [2] established the necessity of combining visual perception with logical reasoning over chart data, yet they often relied on generated questions that lack the ambiguity of natural user intent. Addressing this, [10] curates challenging charts from arXiv papers, revealing a substantial performance gap between proprietary models and human experts, particularly in synthesizing information across complex visual elements. Similarly, [89] and [90] emphasize the need for benchmarks that test reasoning on “wild” charts where underlying data tables are unavailable, forcing models to rely solely on visual encoding.

Beyond isolated charts, modern benchmarks increasingly target long-context document understanding where evidence is scattered across multiple pages and modalities. [97] introduces the complexity of multi-image reasoning within slide decks, requiring models to integrate information across a sequence of visual contexts. This challenge is further amplified in [101], which constructs questions over lengthy PDF documents averaging nearly 50 pages, mandating cross-page retrieval and the synthesis of text, tables, and figures. The [95] dataset formalizes this by requiring joint reasoning over text, tables, and images, demonstrating that single-modality approaches fail significantly on cross-modal questions. Furthermore, [102] and its robust variant [103] raise the stakes by incorporating college-level disciplinary knowledge across diverse fields, testing whether models can perform expert-level perception and deliberate reasoning rather than superficial pattern matching. The emergence of text-rich visual comprehension benchmarks like [104] and [105] further underscores the difficulty of grounding answers in dense textual layouts embedded within images, such as maps and webpages.

In specialized domains, the requirements for precision and factual grounding are paramount. The medical sector sees benchmarks like [106] and [13], which evaluate RAG systems on clinical accuracy and hallucination mitigation using datasets like MIMIC-III. These benchmarks highlight the critical need for retrieving relevant clinical notes and imaging data to support diagnostic generation. Similarly, in the financial domain, [107] focuses on discrete reasoning over semi-structured tables and text in financial reports, a task where standard VQA models lag significantly behind human performance. Despite these advancements, a critical limitation persists: many datasets still rely on

multiple-choice formats or specific answer spans, which may not fully capture the generative fidelity required in open-ended RAG scenarios. Future directions must prioritize benchmarks that evaluate the faithfulness of generated narratives to retrieved visual evidence, moving beyond accuracy metrics to assess the structural coherence and citation fidelity of multimodal outputs, as hinted at by the generative challenges posed in [108] and the retrieval-focused protocols in [75].

5.2 Metrics and Methodologies for System Assessment

Building on the diverse benchmarks and domain-specific challenges outlined previously, the holistic assessment of Multimodal Retrieval-Augmented Generation (MMRAG) systems necessitates a paradigm shift from monolithic accuracy scores to granular, multi-dimensional evaluation frameworks. While existing datasets like CharXiv and MMLongBench-Doc expose performance gaps in complex reasoning, traditional metrics often fail to disentangle retrieval failures from generation hallucinations, obscuring specific bottlenecks in visual reasoning pipelines. To address this, recent methodologies decompose system performance into distinct axes: retrieval relevance, context faithfulness, and visual grounding fidelity. Frameworks such as RAGAS [86] and ARES [109] have pioneered reference-free evaluation by employing lightweight language model judges to score context relevance and answer faithfulness independently. These approaches are critical for chart understanding, where ground-truth answers may be ambiguous or multi-faceted, allowing for the assessment of whether retrieved visual evidence logically supports the generated narrative without relying solely on exact string matching.

A pivotal challenge in this refined evaluation landscape is quantifying visual grounding—the precise alignment between generated textual claims and specific visual elements within a chart or document layout. Standard generation metrics like BLEU or ROUGE are insufficient for capturing spatial coherence or the correct interpretation of graphical marks, a limitation that becomes apparent when moving from benchmark accuracy to real-world application fidelity. Emerging protocols leverage large multimodal models to verify if generated descriptions correspond to the correct regions, such as specific bars in a bar chart or cells in a table. For instance, the Set-of-Mark (SoM) prompting strategy [84] demonstrates that explicit visual markers can significantly enhance grounding capabilities, suggesting that evaluation metrics must increasingly incorporate localization accuracy, such as Intersection over Union (IoU) for bounding boxes or point accuracy for data values. Furthermore, benchmarks like ChartX [90] and MMMU [102] provide rigorous testbeds that demand college-level reasoning, moving beyond simple data extraction to evaluate the system’s ability to synthesize information across heterogeneous modalities, thereby setting the stage for the advanced deployment scenarios discussed next.

Factual consistency and robustness against noisy retrieval contexts represent another critical dimension of assessment, directly addressing the risks highlighted in medical and financial domains. In these high-stakes environments, “context over-reliance”—where a model blindly accepts irrelevant or erroneous retrieved documents—poses severe risks to the integrity of knowledge synthesis. The RULE framework [13] highlights the necessity of calibrating the trade-off between parametric knowledge and retrieved context, proposing metrics that penalize hallucinations induced by noisy evidence. Similarly, studies on robustness reveal that including irrelevant documents can paradoxically improve performance in some text-only settings [48], yet this phenomenon requires careful re-evaluation in multimodal settings where visual clutter and OCR errors are prevalent. Methodologies like eRAG [110] offer a computationally efficient alternative to end-to-end evaluation by assessing the downstream utility of individual retrieved documents, providing a finer-grained diagnosis of retrieval quality essential for reliable system deployment.

Looking forward, the field must converge on unified standards that integrate automated LLM-based judging with human expert validation, particularly for domain-specific applications where nuanced correctness is paramount [106]. Future metrics should also account for the efficiency of the retrieval-generation loop, balancing computational cost against the fidelity of multi-hop reasoning [67]. Ultimately, a comprehensive evaluation ecosystem must not only measure final answer accuracy but also trace the provenance of every claim to its visual and textual source, ensuring transparency and trustworthiness in automated document intelligence and enabling the transition to active, reasoning-driven knowledge synthesis in scientific and financial sectors.

5.3 Applied Deployments in Scientific and Financial Domains

The deployment of Multimodal Retrieval-Augmented Generation (MMRAG) systems in scientific and financial domains represents a paradigm shift from passive data extraction to active, reasoning-driven knowledge synthesis. In scientific research, the primary challenge lies in decoding the dense, heterogeneous information encoded in academic figures and tables, which often contain experimental results critical for meta-analysis but are inaccessible to standard text-based search engines. Recent implementations leverage MMRAG to bridge this gap by retrieving relevant visual evidence from vast corpora such as arXiv and PubMed. For instance, systems utilizing datasets like [111] and [4] enable models to align complex scientific plots with their textual descriptions, facilitating tasks that require cross-referencing visual trends with methodological details. Advanced frameworks such as [89] have demonstrated that instruction-tuning on direct chart images, rather than underlying data tables, significantly improves the fidelity of generated summaries in wild settings, capturing nuanced visual patterns that template-based approaches miss. Furthermore, the integration of chain-of-thought reasoning, as seen in [88], allows these systems to decompose complex inquiries about experimental data into logical steps, retrieving specific chart regions to validate hypotheses before synthesizing a final conclusion.

In the financial sector, the stakes for accuracy and timeliness are equally high, driven by the need to interpret earnings reports, market dashboards, and regulatory filings that blend numerical tables with narrative analysis. Traditional Retrieval-Augmented Generation (RAG) systems often falter here due to the inability of standard chunking strategies to preserve the structural integrity of financial documents. To address this, novel approaches like [35] advocate for element-type-based chunking, which respects the hierarchical layout of reports to ensure that retrieved context maintains its semantic relationship with headers and footnotes. Moreover, the complexity of financial reasoning often demands multi-hop logic across disparate data points, a capability enhanced by hybrid architectures. The [59] framework, for example, integrates Knowledge Graphs with vector retrieval to navigate the intricate relationships between entities in earnings call transcripts, outperforming purely vector-based or graph-based methods in both retrieval precision and answer generation. Similarly, [79] emphasizes the retrieval of textual subgraphs to maintain topological awareness, which is crucial for answering questions involving causal chains in market dynamics.

Despite these advancements, significant challenges persist in both domains. Scientific applications struggle with the generalization to novel chart types and the mitigation of hallucinations when interpreting low-resolution or occluded data, as highlighted by the performance gaps revealed in benchmarks like [10]. In finance, the dynamic nature of market data requires systems capable of incremental learning and real-time updating, an area where continual learning frameworks for generative retrieval are beginning to emerge. Future deployments must increasingly rely on neuro-symbolic reasoning to ensure that generated insights are not only fluent but also mathematically verifiable against the retrieved visual evidence. By combining robust structure-aware indexing with

iterative refinement loops, next-generation MMIRAG systems promise to transform how experts in science and finance interact with information, turning static document repositories into interactive, reasoning-capable knowledge bases.

5.4 Critical Implementations in Healthcare and Legal Sectors

Extending the neuro-symbolic reasoning and structure-aware indexing principles established for scientific and financial domains, the deployment of Multimodal Retrieval-Augmented Generation (MMIRAG) in healthcare and legal sectors represents the frontier of high-stakes automated reasoning. In these fields, the cost of hallucination transcends semantic incoherence, posing risks of clinical malpractice or judicial error; consequently, architectures here must prioritize absolute fidelity, traceability, and privacy above all else. In healthcare, the primary challenge involves synthesizing heterogeneous Electronic Health Records (EHRs), which comprise unstructured clinical notes, time-series physiological data, and diagnostic imagery. Systems such as [50] address this by integrating retrieval from external knowledge graphs with multimodal EHR encoders, effectively grounding predictions in established medical ontologies to mitigate parametric hallucination. Similarly, [112] demonstrates the efficacy of mining semantic interactions between radiology images and reports at scale, enabling disease topic prediction directly from visual inputs supplemented by retrieved textual contexts.

A critical evolution in this domain is the strategic shift from pure generation to retrieval-augmented verification. The [113] exemplifies this by employing a hierarchical policy that dynamically chooses between retrieving template sentences from a database of human-verified reports and generating novel descriptions, thereby ensuring structural robustness. Further advancing this paradigm, [114] introduces fine-grained image-text matching scores to retrieve clinically relevant reports, significantly reducing error severity compared to standard cosine similarity metrics. However, the integration of external knowledge introduces the risk of “context over-reliance,” where models may discard correct internal knowledge in favor of noisy retrieved documents—a challenge echoing the robustness issues noted in earlier sections. To counteract this, [13] proposes a calibrated selection strategy that dynamically adjusts the number of retrieved contexts based on uncertainty estimates, balancing inherent model knowledge with external evidence to maximize factual accuracy.

Parallel complexities define the legal sector, where systems must navigate voluminous regulatory frameworks and interpret dense, layout-rich contracts. The [115] framework illustrates a specialized adaptation for regulatory compliance, tightly coupling retrieval mechanisms with question-answering constraints to ensure generated advice adheres strictly to retrieved statutory guidelines. As observed in financial document processing, the fundamental bottleneck in these applications often resides not in the generative model but in the parsing of source documents. [116] empirically demonstrates that enhancing PDF structure recognition—preserving the spatial relationship between clauses and headers—yields superior retrieval performance compared to standard text-extraction pipelines, which frequently fragment logical legal arguments. Furthermore, the sensitivity of data in both domains necessitates privacy-preserving retrieval mechanisms. While specific cryptographic implementations are reserved for later discussion, the architectural imperative is clear: systems must operate on confidential patient data and proprietary legal briefs without exposing raw visual or textual embeddings during the indexing phase, a constraint that challenges standard dense retrieval paradigms.

Looking forward, the convergence of graph-based retrieval and multimodal understanding offers a promising trajectory for these high-stakes environments. The integration of knowledge graphs,

as seen in [59], allows for multi-hop reasoning across disparate document sections, a capability essential for complex legal case preparation and differential diagnosis. Future research must focus on developing unified evaluation frameworks that quantify not only answer accuracy but also the verifiability of the retrieval chain, ensuring that every generated claim in a medical report or legal brief can be traced back to a specific, authenticated span of visual or textual evidence.

6 Optimization, Robustness, and Ethical Considerations

6.1 Computational Efficiency and Scalable Architectures

The deployment of Multimodal Retrieval-Augmented Generation (MMRAG) systems for chart and document understanding is frequently constrained by the prohibitive computational costs associated with processing high-resolution visual inputs and extensive context windows. As document corpora scale, the quadratic complexity of attention mechanisms in standard transformers becomes a critical bottleneck, necessitating architectures that decouple retrieval efficiency from generative fidelity. A primary avenue for optimization involves extreme context compression, where dense visual embeddings are reinterpreted as direct features within the language model’s latent space, effectively bypassing the need for verbose textual serialization. The [39] framework exemplifies this paradigm, demonstrating that integrating retrieval modality embeddings via a trainable bridge can reduce inference FLOPs by a factor of 3.53 while maintaining performance comparable to uncompressed models. This approach fundamentally shifts the computational burden from the generation phase to the offline indexing stage, enabling scalable deployment on resource-constrained hardware.

Complementary to embedding-level compression, visual token pruning and hierarchical reduction strategies address the redundancy inherent in high-resolution document images. Models such as [31] introduce the H-Reducer module, which merges horizontally adjacent patches to preserve layout information while significantly shortening visual feature sequences. Similarly, [11] employs a Vision Token Merging module that progressively consolidates similar tokens, allowing a 3B parameter model to outperform larger counterparts on chart understanding benchmarks. These methods rely on the premise that not all visual regions contribute equally to semantic reasoning; thus, adaptive filtering mechanisms, as seen in [117], can dynamically discard content-agnostic tokens to optimize both training and inference throughput. Furthermore, [118] advances this by implementing dual-level slimming at both pixel and token stages, adaptively creating compact visual sequences that retain critical structural details necessary for complex layout analysis.

Beyond token reduction, parameter-efficient adaptation offers a scalable pathway for domain specialization without the overhead of full model retraining. By leveraging Low-Rank Adaptation (LoRA) and frozen backbone architectures, systems can tailor generic multimodal models to specific domains like finance or healthcare while preserving the plug-and-play nature of external retrieval indices. The [119] architecture illustrates the efficacy of sparsely activated parameters, where distinct model subsets are specialized for different modalities, achieving performance parity with dense models while activating fewer parameters. This sparse activation strategy is particularly relevant for MMRAG systems that must switch between diverse document types, from scientific figures to legal contracts, without incurring the latency of loading multiple large models. Additionally, generative retrieval paradigms, such as those proposed in [61], offer an alternative to discriminative retrieval by memorizing image identifiers within model parameters, thereby streamlining the retrieval-generation pipeline.

Despite these advancements, a fundamental trade-off remains between compression rates and the preservation of fine-grained numerical data essential for chart interpretation. While methods like

[120] explore frequency-domain processing to capture information with fewer tokens, the risk of losing subtle graphical marks persists. Future research must therefore focus on dynamic, query-aware compression mechanisms that selectively retain high-frequency visual details based on the specific reasoning requirements of the user query, ensuring that scalability does not come at the expense of factual accuracy in data-intensive domains.

6.2 Robustness Against Noise, Perturbations, and Structural Variability

While the subsequent sections address the computational efficiency and privacy imperatives of deploying Multimodal Retrieval-Augmented Generation (MMRAG) systems, the foundational prerequisite for any practical application is robustness against the stochastic noise inherent in real-world chart and document environments. Before optimizing for speed or securing data, systems must first maintain retrieval fidelity and generative faithfulness when input distributions deviate significantly from idealized training corpora due to low-resolution scans, OCR inaccuracies, or intentional adversarial perturbations. Real-world documents frequently exhibit geometric distortions, missing modalities, or layout anomalies that can sever the semantic links between visual marks and textual descriptions, leading to catastrophic retrieval failures if not explicitly addressed.

To mitigate these vulnerabilities, recent architectures have shifted from brittle, sequential pipelines to robust, joint representation learning frameworks. A critical line of defense involves adversarial training strategies that harden vision-language encoders against intentional manipulations. By modeling the joint distribution of heterogeneous modalities, approaches like [19] employ cross-modal adversarial mechanisms to learn discriminative common representations that are invariant to modality-specific noise. Similarly, [53] demonstrates that integrating intra- and inter-modal similarity preservation objectives enables models to learn generic visual features that remain stable even under significant domain shifts or corruption. These methods effectively transform the retrieval space into a manifold where semantically similar documents cluster tightly despite surface-level perturbations, establishing the stability required for the efficiency optimizations discussed later.

Beyond adversarial robustness, handling the inherent noise in optical character recognition (OCR) and layout extraction remains paramount, as traditional systems relying on perfect text serialization fail when outputs are noisy or misaligned. In response, error-tolerant indexing paradigms have emerged that decouple retrieval from precise text extraction. The [121] benchmark highlights the necessity of models capable of operating on noisy, scanned forms where text detection and recognition are imperfect. Addressing this, [122] proposes a unified network where text reading and information extraction reinforce each other, allowing semantic context to correct low-confidence OCR tokens during the retrieval phase. Furthermore, [16] introduces robust layout-aware mechanisms that leverage proper reading orders via Augmented XY Cut, mitigating the impact of noisy positional embeddings derived from flawed OCR tools. For tables specifically, [123] utilizes spatial CNNs to propagate contextual information, enabling robust structure recognition even in geometrically distorted or curved tables where standard grid assumptions fail.

Structural variability and missing modalities present another layer of complexity, particularly when documents are incomplete or layouts vary wildly, necessitating approaches that ensure continuity even when data streams are interrupted. Modality-invariant representation learning offers a solution by enforcing consistency across partial observations. [124] introduces modality-missing-aware prompts that allow transformers to handle general missing-modality cases with minimal parameter overhead, ensuring retrieval continues even when visual or textual channels are degraded. Complementing this, [32] exploits positional, textual, and visual information through cross-modal

pre-training, creating representations that are resilient to the absence of specific modalities. In the realm of generative retrieval, where models predict document identifiers directly, [65] addresses the brittleness of static indices by employing memory-augmented learning to adapt to new, potentially noisy document structures without forgetting prior knowledge.

Despite these advancements in establishing a noise-tolerant foundation, a trade-off persists between robustness and precision; overly agnostic models may retrieve irrelevant documents that share noise characteristics with the query, potentially inflating the computational costs analyzed in the following section. Future research must therefore focus on dynamic noise estimation, where the retrieval mechanism adaptively weights visual versus textual evidence based on real-time confidence scores. Integrating uncertainty quantification, as suggested by probabilistic embedding frameworks, could further refine this balance, ensuring that MMRAG systems remain trustworthy assistants even in the most chaotic document environments, thereby setting the stage for the subsequent discussions on scaling these robust models efficiently and securely.

6.3 Privacy-Preserving Retrieval and Data Sovereignty

The deployment of Multimodal Retrieval-Augmented Generation (MMRAG) in high-stakes domains such as healthcare, finance, and law necessitates rigorous privacy-preserving mechanisms that safeguard sensitive visual and textual data throughout the retrieval-generation pipeline. Unlike unimodal systems, MMRAG architectures face unique vulnerabilities where confidential information may leak through visual embeddings, layout structures, or generated summaries that inadvertently reconstruct proprietary charts or patient records. To address these challenges, emerging frameworks integrate cryptographic protocols with algorithmic perturbations to ensure data sovereignty without compromising retrieval utility.

A primary strategy involves the application of Differential Privacy (DP) to the retrieval process, providing provable guarantees against the leakage of individual document instances. By injecting calibrated noise into the embedding space or utilizing synthetic private queries, systems can prevent adversaries from inferring sensitive attributes of the source corpus. This approach is particularly critical when dealing with heterogeneous data where visual marks in charts may encode sensitive numerical trends [41; 4]. Complementing this, encrypted domain search techniques enable similarity matching over encrypted image embeddings and text vectors, allowing retrieval from private databases without decrypting content during indexing or query phases. Such schemes mitigate the risk of exposure to external attackers and ensure that the generative model only accesses information on a need-to-know basis, aligning with strict regulatory compliance standards [58; 125].

Furthermore, the issue of copyright protection and “near-access freeness” has gained prominence as MMRAG systems risk reproducing identifiable features or proprietary layouts from retrieved sources. Methods such as Copy-Protected Generation (CPR) aim to bound the information extractable from private images, ensuring that synthesized outputs do not violate intellectual property rights or reveal confidential visual structures [126; 127]. This is essential in financial intelligence, where the reproduction of a specific earnings dashboard layout could constitute a breach of non-disclosure agreements. The integration of these protective measures often requires a trade-off between privacy strength and retrieval precision; excessive noise or aggressive encryption can degrade the semantic alignment necessary for accurate cross-modal reasoning [21; 20].

Recent advancements also explore the synergy between retrieval augmentation and secure multi-party computation to facilitate collaborative analysis of sensitive document collections. While traditional RAG systems like those surveyed in [128; 129] focus on accuracy and latency, privacy-centric

adaptations must prioritize data minimization and access control. The challenge is exacerbated in multimodal settings where visual tokens may retain latent information even after textual abstraction [130; 39]. Future research directions should focus on developing lightweight, modality-aware privacy filters that operate at the token level, dynamically adjusting protection mechanisms based on the sensitivity of the retrieved chart regions or document sections. Additionally, the formulation of standardized benchmarks for evaluating privacy leakage in multimodal generation remains an open frontier, crucial for validating the robustness of these systems against sophisticated inference attacks [86; 131]. Ultimately, achieving true data sovereignty in MMRAG requires a holistic rethinking of the architecture, where privacy is not an add-on but a foundational constraint guiding the design of retrieval indices, fusion modules, and generative decoders [80; 76].

6.4 Ethical Alignment, Bias Mitigation, and Fairness in Visual Reasoning

Building upon the foundational requirement for privacy and data sovereignty, the deployment of Multimodal Retrieval-Augmented Generation (MMRAG) in chart and document understanding introduces profound ethical challenges, primarily concerning the propagation of societal biases inherent in visual corpora and the potential for generative hallucinations to distort data-driven narratives. Unlike unimodal text systems, MMRAG must contend with biases embedded not only in linguistic descriptions but also in the visual encoding of data, such as demographic representations in icons or the selective highlighting of trends in financial charts. Recent evaluations reveal that large multimodal models often perpetuate gender and skin tone biases learned from web-scale image-text pairs, leading to stereotypical associations in generated interpretations [132]. This issue is exacerbated in retrieval settings where the underlying knowledge base may overrepresent certain domains or perspectives, as seen in benchmarks like MMBias and FAIntbench which quantify such disparities across chart and document datasets.

Mitigating these biases requires a shift from post-hoc filtering to integrated alignment strategies that operate across the retrieval and generation pipeline, complementing the cryptographic and algorithmic protections discussed previously. One promising direction involves inclusive generation frameworks that leverage reference images to enforce uniform attribute distribution, ensuring that generated visual explanations do not default to majority-class representations [133]. In the context of factual accuracy, the risk of “context over-reliance” poses a significant ethical threat; models may uncritically adopt erroneous or biased retrieved evidence, overriding their internal parametric knowledge. To counter this, calibrated risk control mechanisms have been proposed to dynamically adjust the weight of retrieved context based on uncertainty estimates, thereby preventing the amplification of misinformation [13]. Furthermore, factually augmented reinforcement learning from human feedback (RLHF) has demonstrated efficacy in aligning model outputs with ground-truth visual evidence, significantly reducing hallucination rates in complex reasoning tasks [87].

Transparency and explainability are equally critical for establishing trust in automated visual reasoning, serving as the logical counterpart to the access controls required for privacy. Users must be able to verify the provenance of generated claims, particularly in high-stakes domains like healthcare and finance. Advanced visual grounding techniques now enable the generation of explicit rationales, such as heatmaps or bounding box references, which link textual assertions to specific regions within retrieved charts or documents [134]. This fine-grained attribution allows for the detection of spurious correlations and ensures that the model’s reasoning process is interpretable. However, current approaches often struggle with compositional reasoning, where models fail to correctly associate attributes with objects in complex layouts, a limitation highlighted by benchmarks designed to test localized attribute composition [135].

Future research must address the dual challenge of scaling bias mitigation techniques to massive, heterogeneous document collections while preserving the nuanced semantic relationships required for expert-level reasoning. The development of robust evaluation protocols that go beyond accuracy metrics to assess fairness, social impact, and reasoning faithfulness is imperative. As MMRAG systems increasingly mediate human interaction with data, ensuring their ethical alignment is not merely a technical optimization but a fundamental prerequisite for their deployment in society, bridging the gap between secure data handling and trustworthy AI behavior.

7 Conclusion

The trajectory of Multimodal Retrieval-Augmented Generation (MMRAG) for chart and document understanding has witnessed a paradigmatic shift from static, parameter-heavy models to dynamic, modular architectures capable of leveraging non-parametric external memory. This survey has elucidated how integrating retrieval mechanisms effectively mitigates the hallucination and knowledge staleness inherent in monolithic Vision-Language Models, particularly when addressing the heterogeneity of visual data structures [128]. The evolution from simple dense vector matching to sophisticated hybrid frameworks underscores a critical realization: faithful generation in visually rich domains demands not merely semantic alignment but precise structural grounding. Approaches like [16] and [34] have demonstrated that preserving 2D layout coherence and modeling relational dependencies via graph structures are indispensable for accurate retrieval, surpassing methods that rely solely on lossy text extraction or flat sequence representations. Furthermore, the emergence of generative retrieval paradigms, such as those proposed in [18], offers a promising alternative to traditional bi-encoder architectures by directly predicting visual tokens, thereby bridging the modality gap more seamlessly.

Despite these architectural advancements, significant gaps persist in the current landscape. A primary limitation lies in the fragmentation of evaluation standards, where benchmarks often fail to capture the complexity of real-world reasoning. While datasets like [2] and [100] have advanced the field, they frequently rely on synthetic templates that do not fully stress-test a model’s ability to handle the visual noise and structural variability found in natural documents, as highlighted by the performance drops observed in [10]. Moreover, existing systems struggle with long-context understanding across multi-page documents, where evidence is scattered across disparate visual and textual modalities. The [101] benchmark reveals that even state-of-the-art models suffer substantial performance degradation when required to synthesize information from extensive visual contexts, indicating a fundamental weakness in current attention mechanisms and context compression strategies like those in [31]. Additionally, the robustness of retrieval against imperfect OCR and adversarial perturbations remains an open challenge, necessitating more error-tolerant indexing mechanisms that can function reliably in noisy environments [80].

Looking forward, the future of MMRAG lies in the development of neuro-symbolic reasoning frameworks that combine the fluency of generative models with the precision of symbolic execution. Integrating program-aided reasoning, as seen in Program-of-Thoughts approaches, with retrieval mechanisms will be crucial for ensuring computational accuracy in complex chart interpretation. We anticipate a move towards active, iterative retrieval systems like [66], where models dynamically decompose queries and refine search criteria through multi-hop reasoning loops rather than relying on single-pass fetching. Furthermore, the construction of specialized multimodal knowledge bases, potentially enhanced by Knowledge Graphs as suggested in [59], will enable more structured and explainable retrieval paths. Finally, addressing ethical concerns and bias in visual corpora, alongside developing privacy-preserving retrieval protocols for sensitive domains like healthcare and finance,

will be essential for the responsible deployment of these technologies. By converging structure-aware representation learning, adaptive retrieval strategies, and rigorous evaluation protocols, the next generation of MMRAG systems will achieve the reliability required for expert-level document and chart understanding.

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